Teaching a Neural Network the Glow of the Solar System

A fast zodiacal-light emulator for NASA's SPHEREx mission

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A small neural network reproduces a physics model of the Solar System's dust glow to within 0.015%, up to ~2,000× faster per pointing, and can be extended to every wavelength SPHEREx observes.

Background

SPHEREx is a NASA space telescope, launched in March 2025, that maps the **entire sky in 102 infrared colors** (0.75–5 μm) every six months. Its maps will help explain how the universe began, how galaxies formed, and where water and other ingredients for life are found.

Zodiacal light. The Solar System is filled with a thin cloud of dust from comets and asteroids. Sunlight scattered by this dust, plus the dust's own heat glow, is the **brightest thing in almost every SPHEREx image** (Fig. 1). It changes with where the telescope looks and where Earth is in its orbit, so it must be modeled and removed before faint, distant galaxies can be studied.

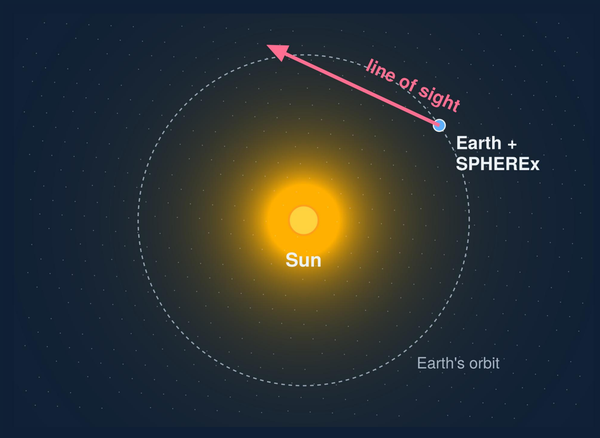


Figure 1. SPHEREx looks out through the Sun's dust cloud (top view, not to scale).

The Kelsall model. Kelsall et al. (1998) built the standard model of this dust from NASA's COBE satellite: a smooth cloud, three dust bands, a ring of dust around the Sun, and a clump that trails Earth. The software **ZodiPy** adds up the glow along each line of sight to predict what a telescope sees (Fig. 2).

Figure 2. The whole sky at 5 μm as seen from SPHEREx (ZodiPy). The glow is brightest toward the Sun and along the plane of the planets. Grey: too close to the Sun to observe.

The problem: computing the model means adding up dust glow along every line of sight, for millions of SPHEREx pointings, at many wavelengths, again and again as the model is compared with real data.

Research objective: to build and test a neural-network **"emulator"** that reproduces ZodiPy's zodiacal light accurately (well under 1% error) while running much faster, ready for real SPHEREx data.

Methods

1. Simulate — ZodiPy computes the zodiacal light for 423,607 SPHEREx pointings (140 days, 5 μm)

2. Describe — 9 numbers per pointing: where the telescope looks relative to the Sun and Earth, plus the time of year

3. Train — A neural network (Flux.jl in Julia) learns brightness from geometry; 5 networks are averaged

4. Test — Compare with ZodiPy on pointings the network never saw: random, unseen months, unseen sky

Give the network the right physics. Telling it where Earth is in its orbit (the "season") cut errors about 40–65×.

Test honestly. Data were split by time, by sky region and at random, and many network sizes and training schedules were compared.

Results

0.015% median error on unseen pointings

~2,000× faster per pointing than ZodiPy

64 wavelengths in one network

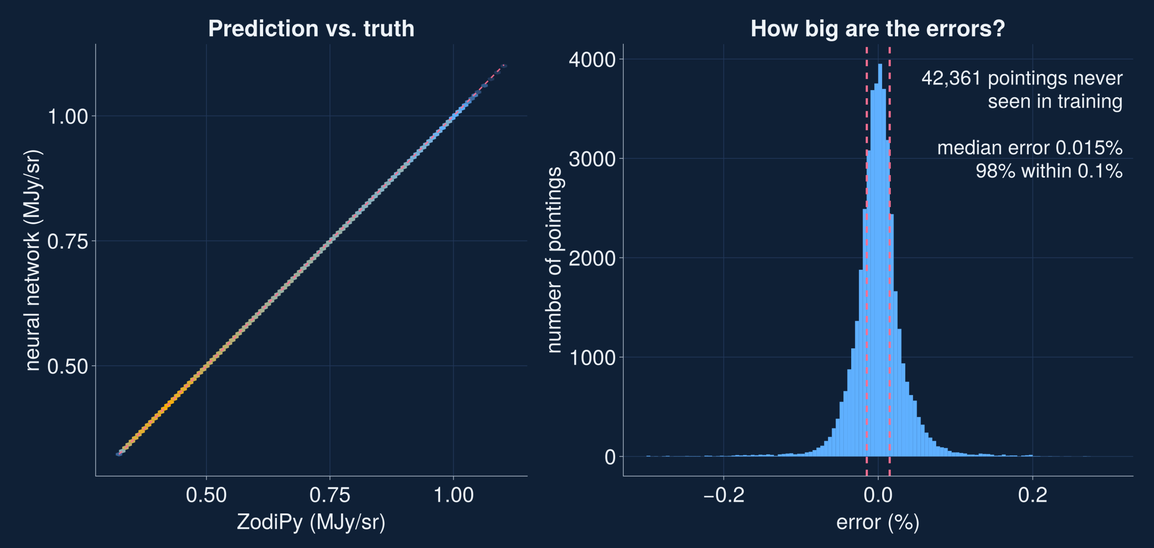


Figure 3. The neural network vs. ZodiPy on 42,361 pointings it never saw in training. Left: every point lies on the 1:1 line. Right: half of the errors are smaller than 0.015%, and 98% are under 0.1%.

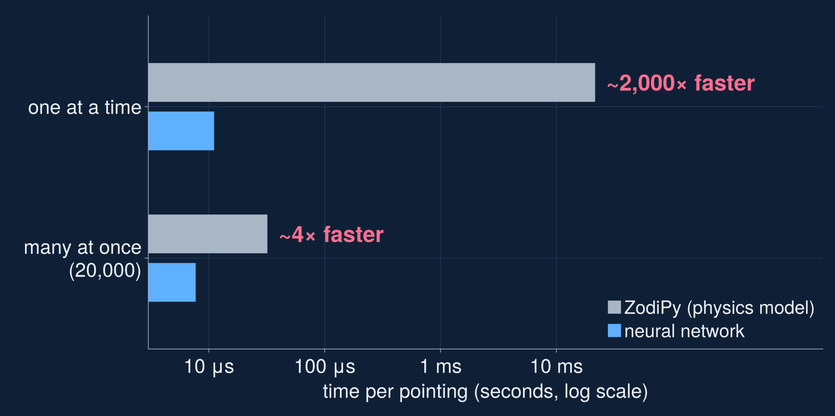


Figure 4. Time per pointing on a laptop (one CPU core, the same 20,000 pointings). One at a time, the network is ~2,000× faster; in large batches, where ZodiPy is efficient, it is ~4× faster.

Looking forward: every SPHEREx wavelength. Real SPHEREx images cover many wavelengths, so wavelength becomes one more input. A first version covering 1.25–5 μm already reaches ~0.1% median error.

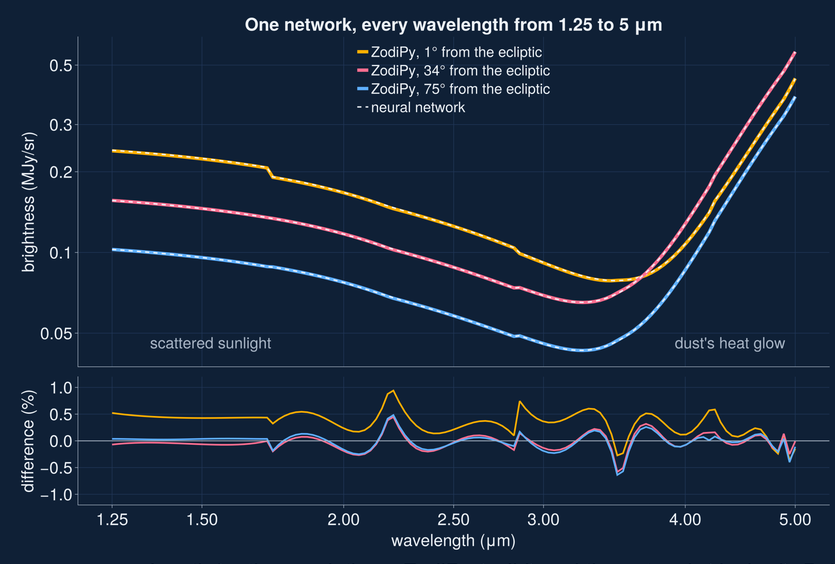


Figure 5. Brightness vs. wavelength for three pointings: ZodiPy (solid) and the network (dashed). Below 3 μm we see scattered sunlight; above, the dust's heat. Bottom: the differences stay under 1%.

Next Steps

✓ DONE Validate — Accuracy, speed, unseen months and sky regions, all on simulated data

IN PROGRESS Refine — Longer training; all SPHEREx wavelengths (0.75–5 μm)

NEXT Scale up — Move training to a computing cluster

NEXT Real data — Apply the emulator to real SPHEREx images

All results so far are preliminary and use simulated data. Next, the model is fully validated and then applied to real SPHEREx data.

Acknowledgements

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References

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